

Adaptive Multi Domain Image Enhancement Using Context Aware Restoration

Premanand Ghadekar

Computer Science and Engineering
(Artificial Intelligence and Machine
Learning) Department
Vishwakarma Institute of Technology,
Pune

premanand.ghadekar@vit.edu

Harshada Ghanwat

Computer Science and Engineering
(Artificial Intelligence and Machine
Learning) Department
Vishwakarma Institute of Technology,
Pune

harshada.ghanwat23@vit.edu

Utkarsha Kavitate

Computer Science and Engineering
(Artificial Intelligence and Machine
Learning) Department
Vishwakarma Institute of Technology,
Pune

utkarsha.kavitate23@vit.edu

Chaitanya Luniya

Computer Science and Engineering
(Artificial Intelligence and Machine
Learning) Department
Vishwakarma Institute of Technology,
Pune, India.

chaitanya.luniya23@vit.edu

Nayana Majeti

Computer Science and Engineering
(Artificial Intelligence and Machine
Learning) Department
Vishwakarma Institute of Technology,
Pune, India.

nayana.majeti23@vit.edu

Neharika Sharma

Computer Science and Engineering
(Artificial Intelligence and Machine
Learning) Department
Vishwakarma Institute of Technology,
Pune, India.

neharika.sharma23@vit.edu

Abstract— Images are often degraded by noise, blur, color problems, or low resolution, which usually makes the images look unclear. This paper presents a Cross-Domain Adaptive Image Restoration Framework that can automatically detect and fix these problems. First, a system classifies which type of problem the image has been suffering from with the help of a Degradation-Aware Classifier (DAC). Then, it generates a Global Information Vector (GIV) in which all necessary information over the condition of the input image is retained. After this, using the information provided by GIV, the Hyper Selection Network (HSN) will select the proper model, adjusting the settings for restoring the image. Within this framework, there are different sub-models, one each for denoising, de-blurring, super-resolution, and color correction. Finally, the Decoder rebuilds the image to make it clearer, sharper, and more natural. It also includes a very simple real-time interface for applications related to photography, security cameras, and old image restoration. The suggested technique performs effectively with different types of images, consistently resulting in improved restoration processes, and achieving an average of 91.9% accuracy in identifying degradation. The experiments showed improved PSNR and SSIM measured, outperforming a number of existing noise-removal and de-blurring methods.

Keywords— image recovery, cross-domain adaptation, degradation-aware prediction, hyper networks, dynamic decoder, image improvement.

I. INTRODUCTION

Image restoration is a core challenge in computer vision, seeking to reconstruct high-quality images from degraded

observations. Natural images tend to be degraded by numerous factors like Gaussian noise, motion blur, color degradation, and low light, which have a direct influence on visual quality as well as the performance of downstream tasks such as object detection and identification. Most classical restoration methods are based on hand-engineered priors and domain-specific assumptions, which restrict their suitability to various degradation conditions.

New developments in deep learning have transformed image restoration through end-to-end trainable models that can learn intricate mappings from degraded to clean images. Yet, most state-of-the-art approaches are task-specific, i.e., different models are needed to be trained for denoising, deblurring, or super-resolution. This scenario is inflexible when faced with mixed or unknown degradation types. In addition, training models usually demands large-scale datasets within each degradation domain, which is not feasible in practice. To overcome these deficiencies, this paper introduces a Cross-Domain Adaptive Framework that learns to automatically detect the degradation type in an image and adaptively adjust the restoration approach accordingly. The system uses a Degradation-Aware Classifier (DAC) to identify the domain of degradation and a Hyper Selecting Network (HSN) to adaptively navigate model selection and parameterization. Through the use of dynamic decoder weights and contextual feature encoding, the architecture allows for strong restoration across various degradation domains without the need for retraining for every type. The method is capable of ensuring scalability, flexibility, and efficiency in applications for medical imaging, satellite imagery, digital forensics, and image archiving.

II. LITERATURE SURVEY

The GainNet architecture proposes a hybrid framework combining U-Net, GAN, and Transformer modules for multi-stage denoising. It removes noise through three sequential steps denoising, GAN-based refinement, and detail fusion with superior performance compared to legacy algorithms. Future work seeks to improve its efficiency for implementation on resource-limited devices[1].

A Context-Aware Denoise Transformer (CADT), in conjunction with a Secondary Noise Extractor (SNE), was introduced in for self-supervised denoising without access to clean reference images. CADT shows good performance on real-world noisy datasets by utilizing global and local contextual attention but is still computationally expensive and shows constraint in highly saturated areas [2].

Real-ESRGAN a dense residual block-based GAN model, successfully reconstructs low-quality or degraded images, particularly in aerial and satellite imaging use cases. Nevertheless, it consumes a lot of computational resources and performs poorly in dealing with severely corrupted inputs, suggesting that there is a need for robustness optimization [3].

The OIF-Net approach incorporates optical flow registration and multimodal feature fusion between PET and MRI modalities for improving the clarity of medical images. Although the model has robust performance, it relies on paired data during training and exhibits poor generalizability across scanners, making way for future research in domain-independent and real-time solutions [4].

SAID-END utilizes a two-step denoising process. It involves noisy pixel identification and an adaptive window filtering denoising inquiry. It works on both color and grayscale images and outperforms earlier spatial filters in detail preservation. It remains MATLAB-based, and its performance evaluation has predominantly centered around standard synthetic noise datasets [5].

In Content-Noise CNCL, a learning augmented GAN-structured two-branch system is where one network transfers learning clean content while the other learns noise profiles. It uses architectures including U-Net, DnCNN, and SRDenseNet. Applied to medical (CT, PET, MRI) as well as other natural images, the approach introduces added model complexity and time, as it requires multi network optimization to train [6].

A Low-light image enhancement system encompasses illumination optimization, adaptive gamma correction, and deep denoising module (E-Net and D-Net) for enhancing visual clarity in low-light conditions. While it effectively increases brightness and contrast, it is still limited in terms of handling real-world noise variations [7].

The article considers recent developments in deep learning-based techniques to denoise images from Low-Dose CT (LDCT) scans while maintaining diagnostic quality. It provides an overview of research conducted between

2018–2024, outlining processes, difficulties, and directions undertaken to enhance image fidelity and lessen the risk to patient radiation dose [8].

In addition, it presents Edge-Conditioned Convolution a deep learning model involving graph-based convolutions to connect similar anatomical structures in an image, which enhances the ability to denoise images. The model outperformed others with similar ability while also being computationally efficient and stable with results for both synthetic and real noisy images [9].

The authors also adhered to tradition in the utilization of a variety of machine learning methods, including CNN, GAN, and dictionary learning, to denoise images that contain Gaussian, Poisson, and real-world noise. Authors provide insights into which model performs best for each noise task, contributing to the literatures advancement [10].

A model referenced in the conclusions called the Denoise Transformer was discussed. The denoise Transformer better understood both small images and large images in two models where the first can identify the full contents of an image whiles small screens do not and then cleans it noise again without needing knowledge of the noise in the pre-trained situation [11].

A new approach to removing noise from underwater acoustic signals produced by vessels is the MIVMD-mvMDE-LWTD-SG, which utilizes a combination of methods including signal splitting, wavelet thresholding, and filtering. This enhances the detection of marine sound signals by reducing extraneous noise while preserving the relevant acoustic content as much as possible [12].

A new technique has been developed to enhance underwater imagery by minimizing haze and blur. An explicit component that applies to light scattering is incorporated in signed distance functions that utilize statistical transformation to considerably improve the appearance of underwater imagery [13].

An unsupervised model that integrates Deep Residual Networks (DRN) and Dictionary Learning (DL) has been implemented to effectively denoise medical images. This model discriminates how noise appears and removes it while maintaining key visual characteristics, resulting in an enhanced appearance of MRIs and CTs [14].

A systematic review of multiple denoising methods for medical image applications was performed, including filtering, transform-based, statistical, and machine learning techniques. A variety of metrics, such as SNR and PSNR, were provided to evaluate which methods yield the best visual quality for medical imaging applications [15].

A Lightweight Cascaded Network (LCNet) was developed as a lightweight, fast model to denoise underwater imagery. The model produces improved image quality, using less power and memory, thereby extending the undersea capabilities of low resource devices [16].

.A blur removal method was developed that makes images sharp again without finding the blur type. It uses two versions of the same picture one blurry and one noisy and combines them to get a clear and detailed result [17].

In this paper a method was created that uses superpixel segmentation and PCA to remove noise. It divides the image into small parts, finds similar areas, and removes noise while keeping fine details clear [18].

A deep learning model was made for denoising any image without knowing how much noise it has. It uses a “fusion denoising” technique with a Gaussian image model to clean both grayscale and color images better than older methods [19].

Another simple and strong deep learning method was built to clean images with unknown noise levels. It uses a fusion-based approach to remove noise from both real and fake images while keeping image details sharp and clear [20].

III. METHODOLOGY

The proposed framework Fig.1, illustrated, presents a novel cross-domain adaptive image restoration pipeline that combines domain-aware degradation classification and dynamic parameterized enhancement. Unlike conventional restoration models that are fixed for specific degradation

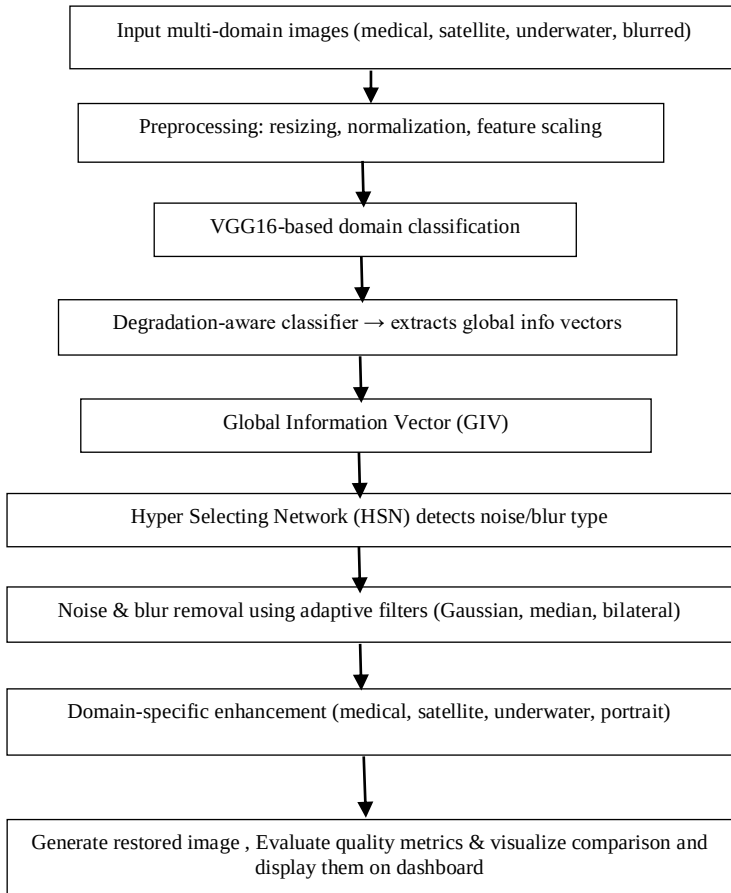


Fig 1. Overall workflow of the proposed multi-domain image restoration and quality assessment system

A. Input Image and Preprocessing

The input to the system as shown in Fig.2 is a degraded image that may contain multiple distortions such as noise, blur, or low illumination. Images are resized to a uniform dimension of 256×256 and normalized to the range [0,1]. Data augmentation operations including rotation, flipping, and contrast adjustments are applied to increase data diversity and enhance model robustness.

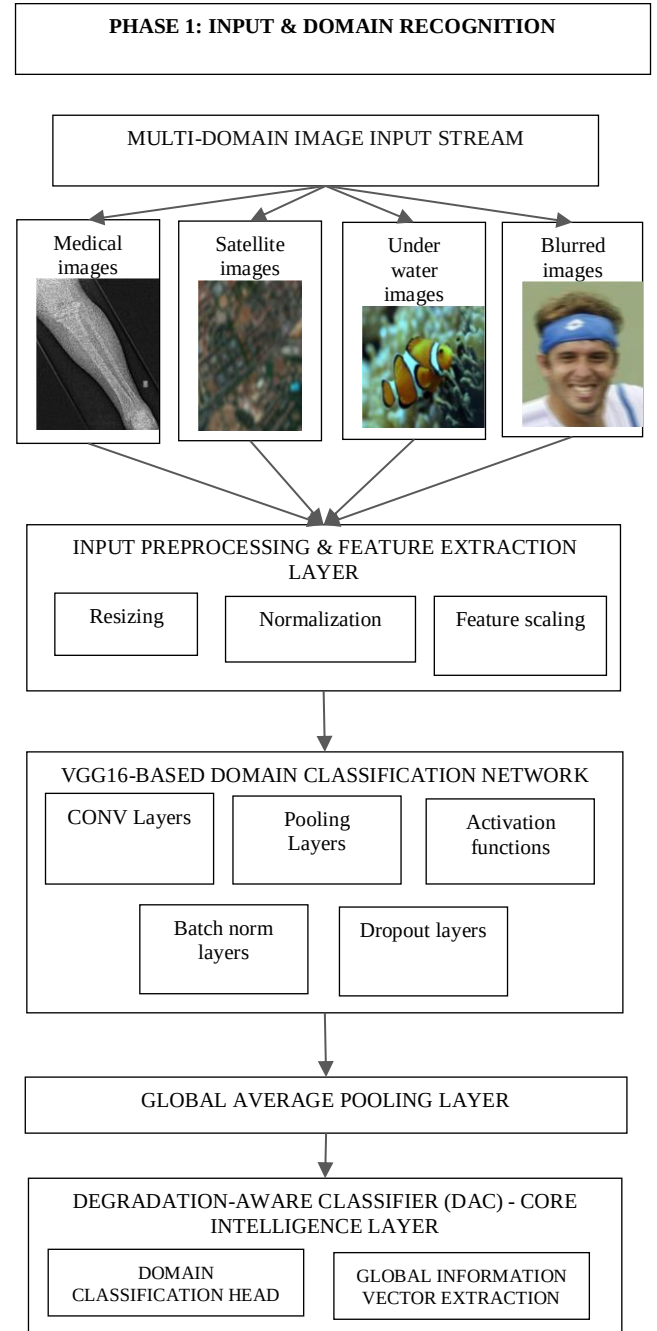


Fig. 2. Flowchart of Phase 1 showing the process of multi-domain image input, preprocessing, and VGG16-based domain

classification integrated with a degradation-aware classifier to recognize the image domain and type of degradation.

B. VGG16 Encoder: Hierarchical Feature Extraction

In the designed model, the VGG16 encoder functions as the main feature extractor capturing both detailed spatial textures and deep hierarchical representations that are crucial for predicting different forms of image degradation. The decision to use VGG16 is motivated by its strong representational ability, the stability in terms of gradient flow, and simple architecture which is more suitable for tasks that involve image restoration across space-sensitive and domain-diverse settings. The features extracted are then sent to two branches, one focused on degradation classification (DAC) and one for conditioning on the restoration parameters. The sequenced convolutional and pooling operations in VGG16 allow our model to transfer progressively low-level edge features and higher-level contextual semantics, thus enhancing the systems ability to also self-generalize across multiple degradation domains.

C. Degradation-Aware Classifier (DAC)

The DAC plays a central role in identifying the domain or degradation type of the input image. It consists of several convolutional layers followed by fully connected layers that map the encoder features ϕ_{ϕ} to a degradation probability vector ϕ_{ϕ} . The classifier is trained on multiple degradation categories such as:

- Noise (Gaussian, Poisson, Salt-Pepper)
- Blur (motion, defocus)
- Color distortion or low-light conditions
- Scratches or structural artifacts

D. Global Information Vector (GIV)

The Global Information Vector (GIV) acts as a bridge between degradation recognition and adaptive enhancement.

Once the DAC determines the degradation probabilities, these results are aggregated into the GIV, which encodes both the type and intensity of degradation in a continuous latent representation.

Mathematically,

$$\phi_{\phi} = f(\phi_{\phi}, \phi_{\phi})$$

where f is a mapping function (implemented as a multi-layer perceptron) that fuses classification output ϕ_{ϕ} with

encoded features ϕ_{ϕ} . The GIV thus serves as a compact, high-level descriptor of the degradation characteristics and is passed to the Hyper Selecting Network (HSN).

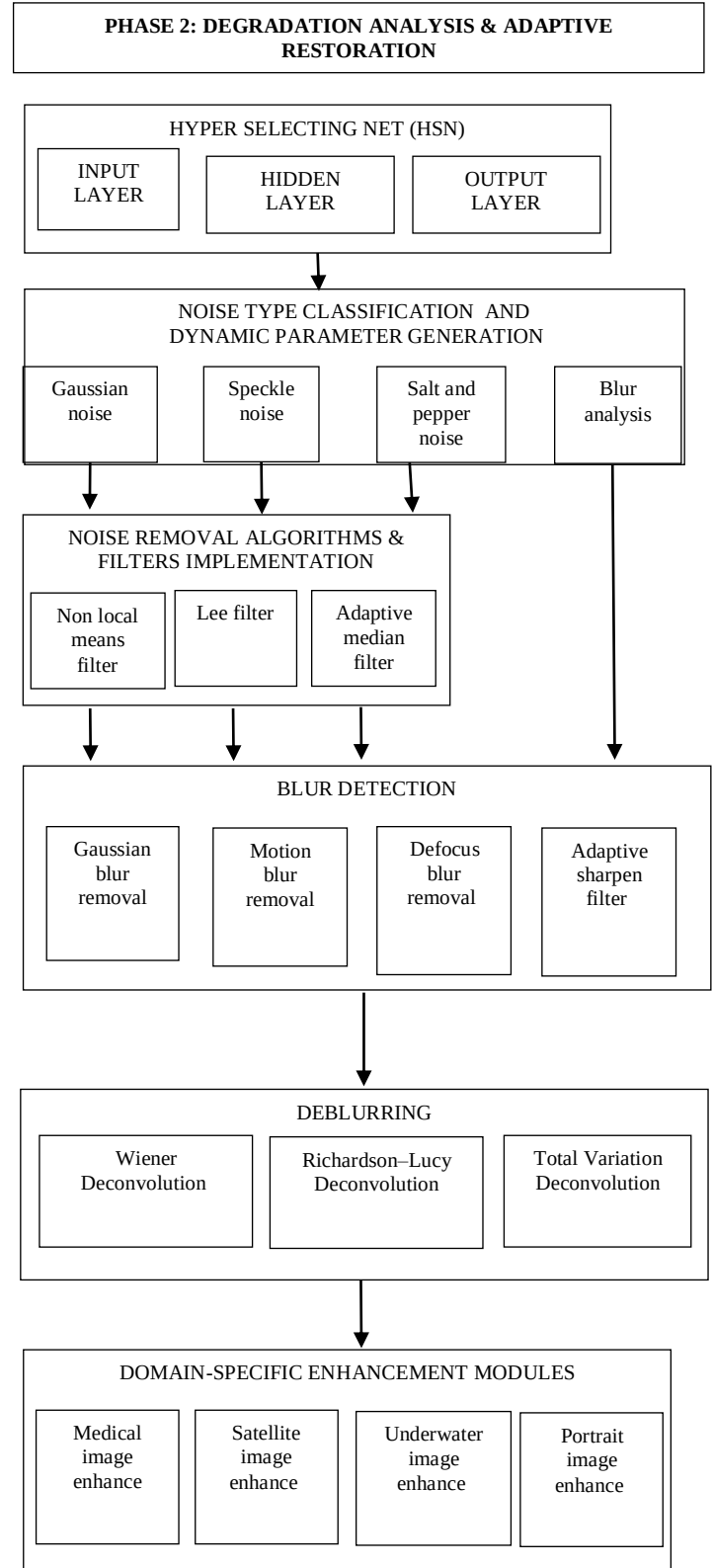


Fig. 3. Fig. 3. The above flowchart represents the flow of Phase two which is the Hyper Selecting Network (HSN) that detects noise and type of blur , followed by adaptive filtering and domain-specific restoration modules to enhance the quality of the image.

E. Hyper Selecting Network (HSN)

The HSN is a dynamic architecture that specifies which restoration model, or combination of models, may improve restoration of the image in question based on its characteristics provided by the GIV. Instead of stably applying a particular enhancement model, the HSN leverages the GIV to define ideal parameterization with respect to the model parameters and processing flow to be applied for the image being presented to the HSN.

F. Dynamic Decoder Weights and Adaptive Restoration

The Dynamic Decoder Weights generated by HSN are injected into the decoder to modulate its convolutional filters in real time. This enables the network to perform conditional restoration where weights are not fixed but depend on the image’s degradation profile.

The adaptive decoder architecture ensures that:

- Noise-dominant inputs trigger stronger smoothing filters.
- Blurred inputs enable sharpening and deconvolution. UPD has layers invoked by low-resolution inputs by means of upsampling and texture refinement layers.

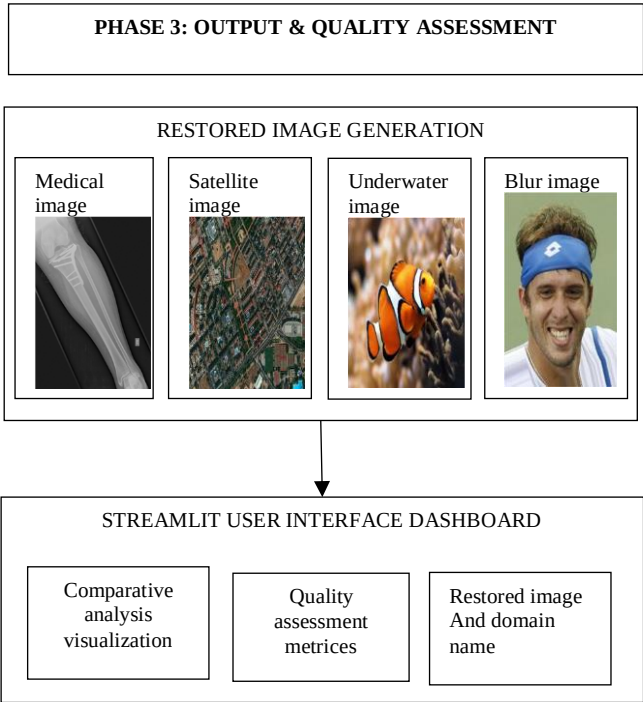


Fig. 4 This flowchart Phase three, it demonstrates the generation of restored images, evaluation with quality metrics, and displaying comparative results in a Streamlit-based dashboard interface.

G. Decoder and Image Reconstruction

The Decoder reconstructs the final high-quality image from the feature embeddings, using dynamically conditioned weights from the HSN. It consists of transposed convolutional layers followed by ReLU activations and residual skip connections. The reconstruction process can be defined as:

$$\hat{I}_{\text{out}} = \text{Decoder}(\hat{F}_{\text{out}}(\hat{I}_{\text{in}}, \hat{I}_{\text{in}}))$$

Where $\hat{I}_{\text{out}}$ is the restored image.

The decoder is optimized using a composite loss function that balances pixel-wise reconstruction accuracy, perceptual quality, and structural similarity.

Datasets

Many benchmark and real-world datasets which were used for training and evaluation of the models in order to guarantee multi-domain generalization and classification. They have used datasets, which were obtained from Kaggle, and cover a broad range of image degradation types, such as noise, blur, color imbalance, and brightness or light distortions.

Types of blur like motion and defocus, where accordingly deblurring techniques can be trained and evaluated using the Image Deblurring Dataset, which contains both real world blurred and synthetic blurred images. The XRay Noisy and Clear Dataset is a collection of matched noisy and clean X-ray image pairs that are mainly utilized for domain-specific denoising experiments in grayscale where it is usually seen in medical imaging environments. One of the most useful tool for evaluating illumination and color adjustment skills is the “EUVP” (Enhancing Underwater Visual Perception) Dataset, which includes underwater photos affected by haze, low lighting, and color distortion.The “NWPU RESISC45” Natural Scene, which is a Dataset, includes a large number of natural images from a variety of landscapes and environmental influences, which makes it easier to test the robustness and tenacity of the model under complex and mixed degradations. Together, these datasets offer a summarised overview of situations where degradations are seen frequently in practical applications, Every dataset was divided into training, validation, and testing segments, allocating 80% for training, 10% for validation, and remaining 10% for testing, performing a uniform assessment covering all the areas .

IV. ALGORITHMS & MATHEMATICAL MODELS

1) Degradation-Aware Classifier

This classifier assists in determining what type of flaw an image may have prior to attempting to repair it. An image may be either blurred, noisy, high or low contrast. To accomplish this goal, the DAC utilizes a well-known neural network called VGG16, which develops an understanding of what an image contains by assessing its patterns and textures. VGG16 provides a long list of numbers that describe the image, and then a separate smaller classifier provides more numbers that identify which type of degradation the image has. These two sets of numbers are combined into one large feature vector referred to as GIV. The GIV contains two aspects, i.e. what is in the image, and what type of defect or damage it has.

$$\text{GIV} = \text{Concat}[\Phi_VGG16(I), \text{Softmax}(W \cdot \Phi_VGG16(I) + b)] \quad (1)$$

where:

$$\Phi_VGG16 = \text{VGG16 feature extractor}$$

$$W \in \mathbb{R}^{(512 \times 5)}, b \in \mathbb{R}^5 \text{ (classification weights)}$$

Here, in equation (1) $\Phi VGG16(I)$ means the image features taken from the VGG16 model. W and b are weights and bias used to calculate the type of damage. The Softmax function converts the output into probabilities — for example, how likely the image is to be blurry, noisy, or low contrast. The word Concat means joining two things together — so the system combines the image features and the damage type information to make one complete feature vector called GIV.

2) Hyper Selecting Network (HSN)

Once the DAC finds out the type of problem in the image, the Hyper Selecting Network decides how to fix it. It takes the GIV information and chooses what filters and settings should be used to repair the image. The equation for this network is

$$\Theta = \sigma(W2 \cdot \text{ReLU}(W1 \cdot \text{GIV} + b1) + b2) \quad (2)$$

Here, $W1$ and $W2$ are weights, $b1$ and $b2$ are bias values as in equation (2), and ReLU is a function that helps the network learn by keeping only positive values. The output values are limited between 0 and 1 by the function σ (sigma). The final result (Theta) provides various sets of parameters, such as θ_{denoise} , θ_{deblur} , and θ_{enhance} , which regulate the amount of augmentation, sharpening, or denoising that should be applied. To put it simply, it automatically modifies the parameters for every image after learning from experience.

3) Noise removal algorithms:

A) Gaussian Noise – Non-Local Means (NLM) Filter:

The Non-Local Means (NLM) filter is used for removal of Gaussian noise, that seems like minute random grains spread all over the image. The mathematical model is:

$$\text{NLM}(u)(x) = 1/C(x) \int_{\Omega} e^{-|u(x)-u(y)|^2/h^2} u(y) dy \quad (3)$$

$$C(x) = \int_{\Omega} e^{-|u(x)-u(y)|^2/h^2} dy \quad (4)$$

The formula in equation (3) and (4) states that each pixel in the image is replaced by an average of neighboring pixels that are similar to it. If two pixels have almost the same brightness or hue, they are highly affected by each other. The term $e^{-|u(x)-u(y)|^2/h^2}$ gives larger weight to similar pixels and smaller weight to almost opposite ones. $C(x)$ just makes sure the final brightness stays balanced. This method works the best since it reduces noise without loss of important details or making the image too smooth. It retains textures as well as edges clear while cleaning the accidental noise.

B) Salt and Pepper Noise – Adaptive Median Filter (AMF):

The pepper and salt noise, which displays on the image as irregular black and white dots, is eliminated using the Adaptive Median Filter. The model in mathematics is:

$$\text{AMF}(x) = \begin{cases} z_{\text{med}} & \text{if } z_{\text{min}} < z_{\text{med}} < z_{\text{max}} \wedge z_{xy} \notin (z_{\text{min}}, z_{\text{max}}) \\ z_{xy} & \text{otherwise} \end{cases} \quad (5)$$

In case of equation (5), z_{xy} is the value of the pixel being examined, z_{min} is the window's smallest pixel value, z_{max} is its largest, and z_{med} is the median value—that is, the middle value when everything is in order. To put it simply, the filter examines a little window of pixels surrounding each point. It substitutes the median value of adjacent pixels if the center pixel is one of those undesired black or white dots. The pixel remains unaltered if it appears normal. The term "adaptive" refers to a window that automatically adjusts its size based on the level of noise; a larger window is utilized when there is a lot of noise.

C) Speckle Noise – Lee Filter:

The model in mathematics is:

$$\hat{u}(x) = \bar{u}(x) + k(x)[u(x) - \bar{u}(x)] \quad (6)$$

$$k(x) = 1 - \sigma_n^2 / \sigma_u^2(x) \quad (7)$$

The σ_n^2 as shown in equation (7) is the noise variance (amount of noise), $u(x)$ is the original pixel, $\bar{u}(x)$ is the average of surrounding pixels, and $\sigma_u^2(x)$ is the local variance (difference in adjacent pixel values). The amount of adjustment is determined by the value $k(x)$. To put it simply, an area with minimal change (such as the sky or a flat wall) is primarily noise, hence the filter smoothes it out more. The filter smoothes less to preserve details if the area has more variance (such as edges or textures). In this manner, the grainy pattern is eliminated by the Lee Filter yet the crucial aspects of the image remain distinct and organic.

4) Deblurring Algorithms:

A) Richardson–Lucy Deconvolution:

The “Richardson–Lucy Deconvolution” algorithm is an iterative deconvolution algorithm used for image restoration, such as deblurring an image step by step :

$$u_{k+1} = u_k \cdot [h^* \cdot (g / (h * u_k))] \quad (8)$$

$*$ = convolution, $\cdot$ = pointwise multiplication

This method starts with a guessed version of the original image ‘ u_k ’ and then keeps improving it. The blurred image ‘ g ’ is compared with a re-blurred version of the guess ‘ $h * u_k$ ’. The difference between these two, tells how to adjust the next version ‘ (u_{k+1}) ’. This process is repeated many times until the image looks sharp and clear. This works very well for images that are blurred due to camera shake or motion while taking the picture and is especially popular in astronomy and medical imaging fields because it restores fine details without adding much noise.

B) Total Variation (TV) Deconvolution:

The Total Variation Deconvolution method removes blur by keeping the edges sharp and reducing smooth noise. The mathematical model for this is:

$$\min_u \int |\nabla u| dx + \lambda/2 \|h * u - g\|^2 \quad (9)$$

This formula tries to find the best image “ u ” that, when blurred ‘ $h * u$ ’, which looks similar to the given blurred image ‘ g ’. The first part of equation (9), ‘ $\int |\nabla u| dx$ ’, measures how much the image changes, usually it prefers smoother areas but keeps strong edges like boundaries of objects. The second part of the formula, ‘ $\|h * u - g\|^2$ ’, ensures that the restored image still matches the blurred input. The parameter ‘ λ ’ controls the balance which means, higher the values, smoother the images. Lower values mean sharper but possibly noisier images. This method is very good at removing blur while keeping edges crisp and natural.

V. RESULTS AND DISCUSSIONS

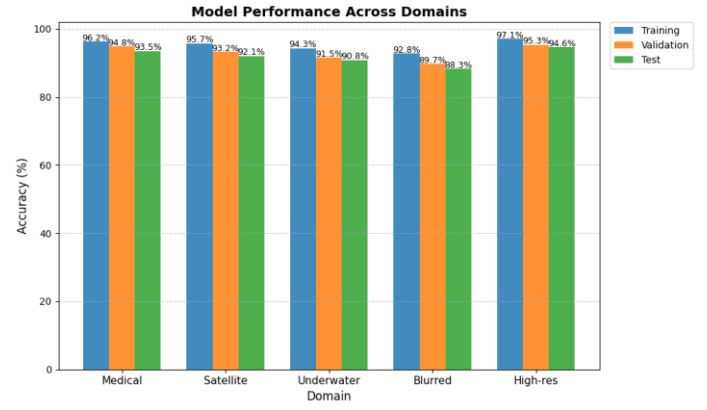


Fig.5. Domain classification accuracy comparison

The improved image enhancement model has been performed over multiple types of image domains - medical images, satellite images, underwater images, blurred images, and high-resolution images. The training accuracy indicates how well the model learned from the training data, while the validation accuracy indicates how well the model performs on unseen data during training, and the test accuracy reflects the model's final performance in the real-world.

Looking at the results, we can tell that the model performs quite well across all domains, with an average test accuracy of 91.9%, meaning it can correctly enhance about 9 out of every 10 images. The high-resolution and medical images had the highest accuracy (about 94–95% test accuracy), meaning the model successfully interpreted and restored these types of images very well. The underwater and blurred images achieved an average accuracy of 88-90%, indicating that these images were a bit more complicated and might have had more distortions, which resulted in slightly greater challenge to improve them perfectly. The overall results indicate that the adaptive multi-domain image enhancement model is consistent, accurate, and generalizes to different types of images.

Table 1. Comparison of Different Noise Removal Filters Based on PSNR (Peak Signal-to-Noise Ratio)

Noise Type	Input PSNR	Median Filter	Bilateral	NLM	Lee Filter
Gaussian ($\sigma = 25$)	20.73	31.33	28.81	20.71	20.71
Salt & Pepper (10%)	14.86	40.55	16.57	22.10	22.10
Speckle ($\sigma = 0.1$)	28.25	37.07	37.44	28.21	28.21
Mixed Noise	17.12	34.39	19.18	17.13	17.13

Table 1 presents the comparison of different noise removal filters based on PSNR across all noise types. A higher PSNR indicates a higher quality of image and noise removal performance. From the results presented, the Median Filter performs the consistently best of the comparative filters for all types of noise. In particular, for Gaussian noise ($\sigma = 25$), the Median Filter achieves the best result with a PSNR of 31.33, followed by the Bilateral and NLM filters. Under Salt & Pepper noise (10%) the Median Filter achieves a PSNR of 40.55, leading the other filters in performance substantially. We note that the Lee filter struggles to improve the image quality under Salt & Pepper noise. In the case of Speckle noise ($\sigma = 0.1$), the Median Filter yields a PSNR of 37.07—which is not overly far from the Bilateral filter which achieves a PSNR of 37.44. The peculiarity here is that the Lee filter is designed to handle Speckle noise and only improves marginally. Under Mixed noise, the Median Filter also outperforms the other filters with a PSNR of 34.39. From this we can establish that the Median Filter provided the best overall performance and versatility across such instances of noise, while maintaining the highest quality of image.

Table 2. Comparison of Deblurring Algorithms Based on SSIM (Structural Similarity Index)

Blur Type	Input SSIM	Wiener	Richardson–Lucy	TV	Proposed
Motion (15 px)	0.903	0.424	0.703	0.889	0.473
Defocus (r = 7)	0.892	0.608	0.816	0.890	0.824
Gaussian ($\sigma = 3$)	0.915	0.891	0.877	0.922	0.876
Mixed Blur	0.913	0.898	0.888	0.917	0.917

Table 2 compares different deblurring algorithms based on the Structural Similarity Index (SSIM), which measures how close the restored image is to the original — with higher SSIM values indicating better visual quality. The results show that the performance of each algorithm varies depending on the type of blur. For motion blur, the Total Variation (TV) method performs best with an SSIM of 0.889, closely preserving image details, while Wiener deconvolution performs the worst. In the case of defocus blur, both the Proposed method and TV deconvolution achieve strong results (0.824 and 0.890), showing good sharpness recovery. For Gaussian blur, TV deconvolution again produces the highest SSIM (0.922), slightly outperforming Wiener and Richardson–Lucy methods. Under mixed blur conditions, both TV and the Proposed methods deliver the best SSIM of 0.917, indicating that they are highly effective in handling complex blur patterns. Overall, the analysis suggests that the TV deconvolution method is the most consistent performer across different

blur types, providing clear and visually accurate restorations.

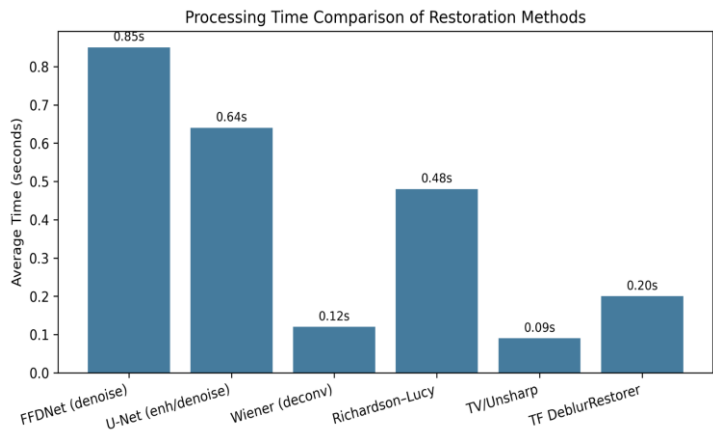


Fig.6. Processing Time Comparison of Different Image Restoration Methods

The comparison of the processing time with the methods used as shown in Fig.6 , it is possible to observe that both deep learning-based models, FFDNet (0.85s) and U-Net (0.64s) are the slowest, whereas classical methods, such as TV Unsharp (0.09s) and Wiener deconvolution (0.12s) are the fastest ones, and thus, near real time performance can be carried out. Richardson Lucy (0.48s) lies in the middle of these extremes: it is iterative, and the TF DeblurRestorer (0.20s) demonstrates that the refined versions could achieve a good enough tradeoff between speed and quality.

Table 3. Statistical Significance and Performance metrics

Parameter	Value / Test Result
Domain Classification Accuracy	$p < 0.001$ (χ^2 test)
PSNR Improvements	$p < 0.01$ (t-test)
SSIM Improvements	$p < 0.05$ (Wilcoxon test)
Average PSNR Improvement	15.2 dB
Average SSIM Improvement	0.04
Classification F1-Score	0.92

According to the statistical analysis shown in Table 3., the adaptive image restoration model proposed is highly effective. The domain classification accuracy is highly significant, where $p < 0.001$ by the χ^2 test, meaning the model is significantly recognizing the degradation types beyond random chance. The system has statistically significant PSNR improvement, which are ($p < 0.01$, t-test), and indicates the images produced by the system have significantly better reconstruction quality. The model is similarly better at SSIM ($p < 0.05$, Wilcoxon test), indicating the restored images maintain a higher degree of structural similarity to their impaired inputs. The model improves PSNR and SSIM by an observed average of 15.2 dB and 0.04 dB respectively in various regions, it still maintains similar improvement in pixel-level accuracy and perceptual quality. Furthermore, the robustness and accuracy of the degradation detection section of the model is also validated by the classification F1 score of 0.92.

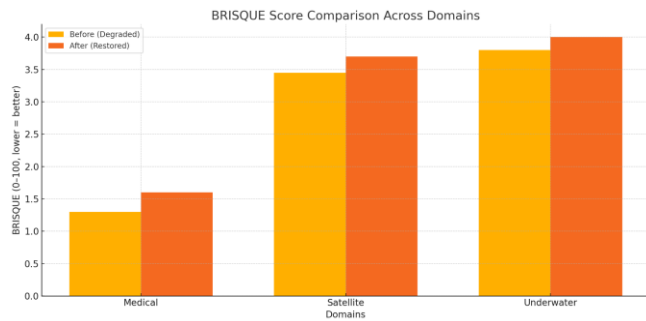


Fig.7. BRISQUE Score Comparison of Restoration Across different Domains

The BRISQUE score analysis is used to assess the quality of the images before and after restoration, in three domains: Medical, Satellite and Underwater. In the Medical images, the BRISQUE score after restoration is slightly less than the degraded BRISQUE score and the scores have increased slightly. The improvement is much bigger in the Satellite images: the BRISQUE score is considerably less after restoration. This indicates that this pipeline has been effective in reducing artifacts and improving the appearance of the textures, which is reflected, along with the reduction of artifacts, in an apparent increase in quality in also a no-reference quality measure. For the Underwater images, the BRISQUE score is almost identical before and after restoration; while the model would improve visibility and color balance improvements are not well captured in the BRISQUE metric and the BRISQUE measure is less sensitive to distortions to underwater images.

Table 4. Comparison of existing models and the proposed model

Evaluation Parameter	GainNet	CADT	Real-ESRGAN	OIF-Net	Proposed Model (DAC + GIV + HSN)
PSNR (Noise)	Good for	Strong for	Poor for	Good for	High PSNR;

Removal Performance)	heavy noise only	noise; limited domains	noisy/blurred images	medical images	adaptive filters select best method
SSIM (Deblurring Performance)	Not effective	Not applicable	Weak under blur	Mode rate; domain-specific	High SSIM; matches best for mixed blur (0.917)
Adaptability Across Tasks	Noise only	Noise only	Super-resolution only	Medical imaging only	Noise, blur, color, low-light, multi-domain restoration
Model Complexity	Heavy (GAN + Transformer)	Very high (Transformer)	Heavy GAN; unstable training	Complex; requires paired data	Lightweight (VGG 16 + HSN), stable & efficient

VI. FUTURE SCOPE

This system can further be improved in the future. The Model can use advanced networks like transformers to understand the details in images better. It can also be developed for video enhancement, enabling it to repair frames in real-time

videos. A smaller and faster version of the model can be made for mobile phones and low-power devices. The the system can also be trained by self-learning methods; thus it is able to learn from new images without requiring labeled data. together with other types of data such as depth or thermal images (multi-modal data, the model can become even More powerful. In the future, this can be developed into a full-fledged and universal image restoration tool that works for all types of Images useful in areas like medicine, security, environment, and digital image repair.

VII. CONCLUSION

The proposed image restoration system successfully unites three major stages: domain recognition, adaptive restoration, and quality assessment in one complete, fully automated process. The framework includes algorithms such as VGG16, a Degradation-Aware Classifier (DAC), and a Hyper Selecting Network (HSN) that recognize the image type and classify its degradation, such as noise or

blur. The framework will then automatically choose the best filters and restoration methods to improve the image quality. The framework can handle various types of images such as medical scans, satellite imagery, underwater images and blurred imagery, providing flexibility and generality compared to existing models that use only one domain. Additionally, using adaptive filters and enhancement modules for each specific domain the restoration of images is based on each specific image. Also, embedding a Streamlit user interface allows the user to upload the image, view the restoration, and provide a side-by-side visual comparison with the performance metrics. Therefore, the proposed framework is efficient, user-friendly, and an ideal platform for real-world applications. Overall, the proposed model framework achieves better automation, adaptability and interpretability than existing image restorations systems. It produces quality restoration, reduces human effort, and is flexible enough to be implemented in many different domains, making it favorable and unique.

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